

The importance of data structure in statistical analysis of dendritic spine morphology

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The inspiration



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Short communication

The importance of data structure in statistical analysis of dendritic spine morphology

Veerle Paternoster^{a b c d} , Anto P. Rajkumar^{a b d e f} , Jens Randel Nyengaard^g ,
Anders Dupont Børglum^{a b d} , Jakob Grove^{a b d h 1} , Jane Hvarregaard Christensen^{a b d 1}

Highlights

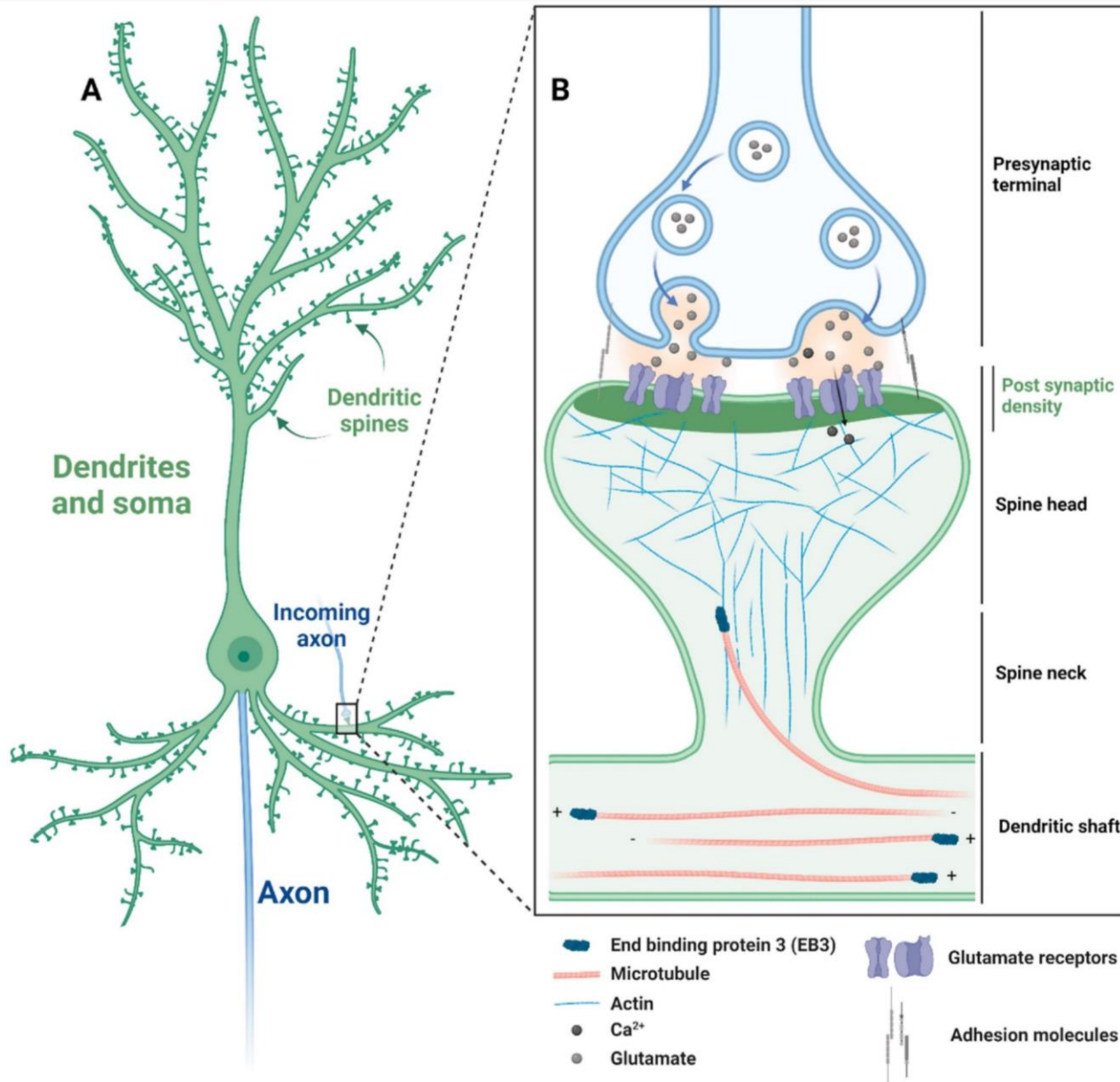
- Data collection for dendritic spine morphology often results in multilevel data structure.
- The need for implementation of data structure in statistical models is highlighted.
- The use of mixed-effects models to study dendritic spine morphology is proposed.
- Mixed-effects models are superior to commonly used statistical methods for studying dendritic spine morphology.

Paternoster et al., 2018

Questions

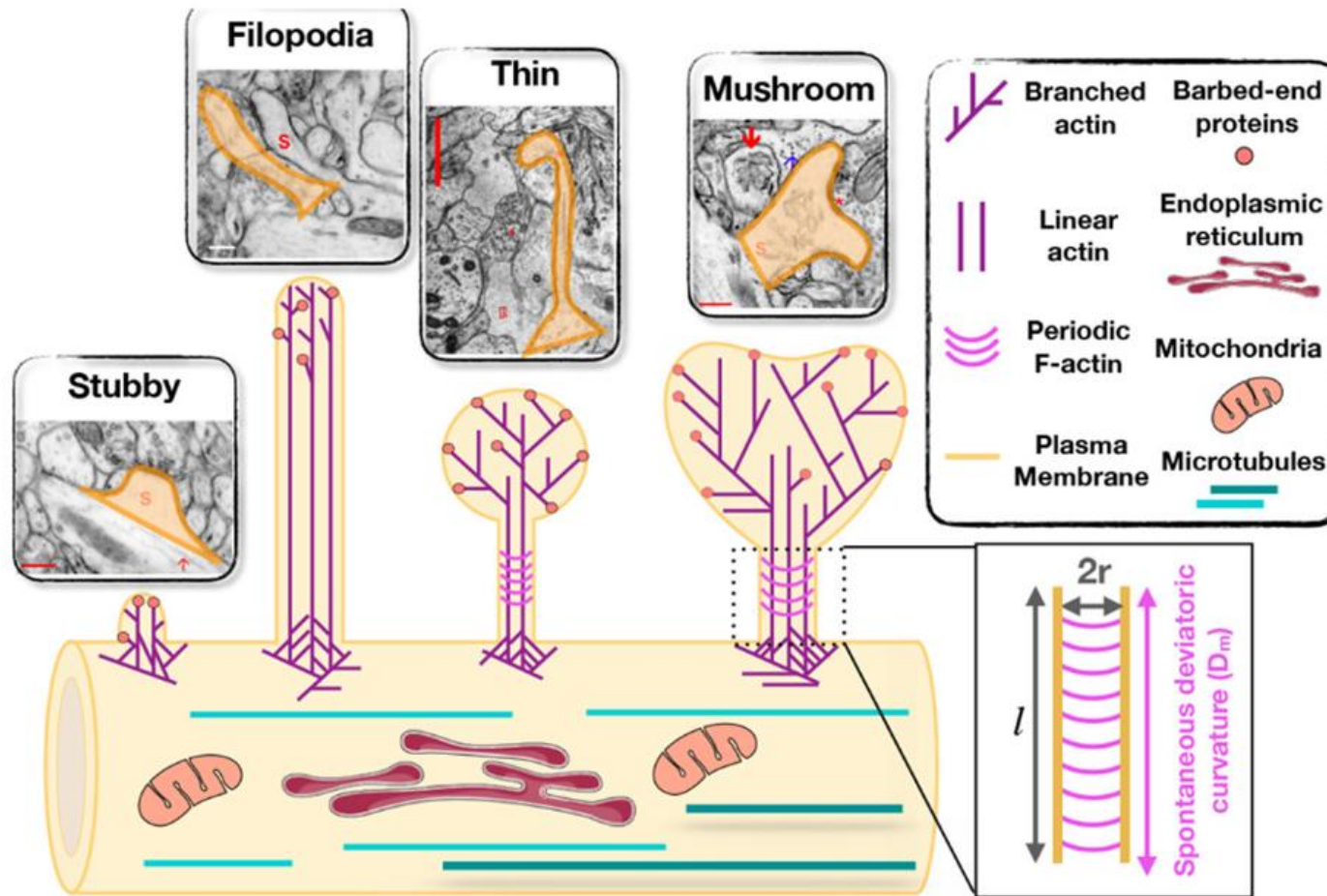
- Why linear mixed-effect models are important?
- How to apply linear mixed-effect models in a neuroscience experiment?

What are dendritic spines?



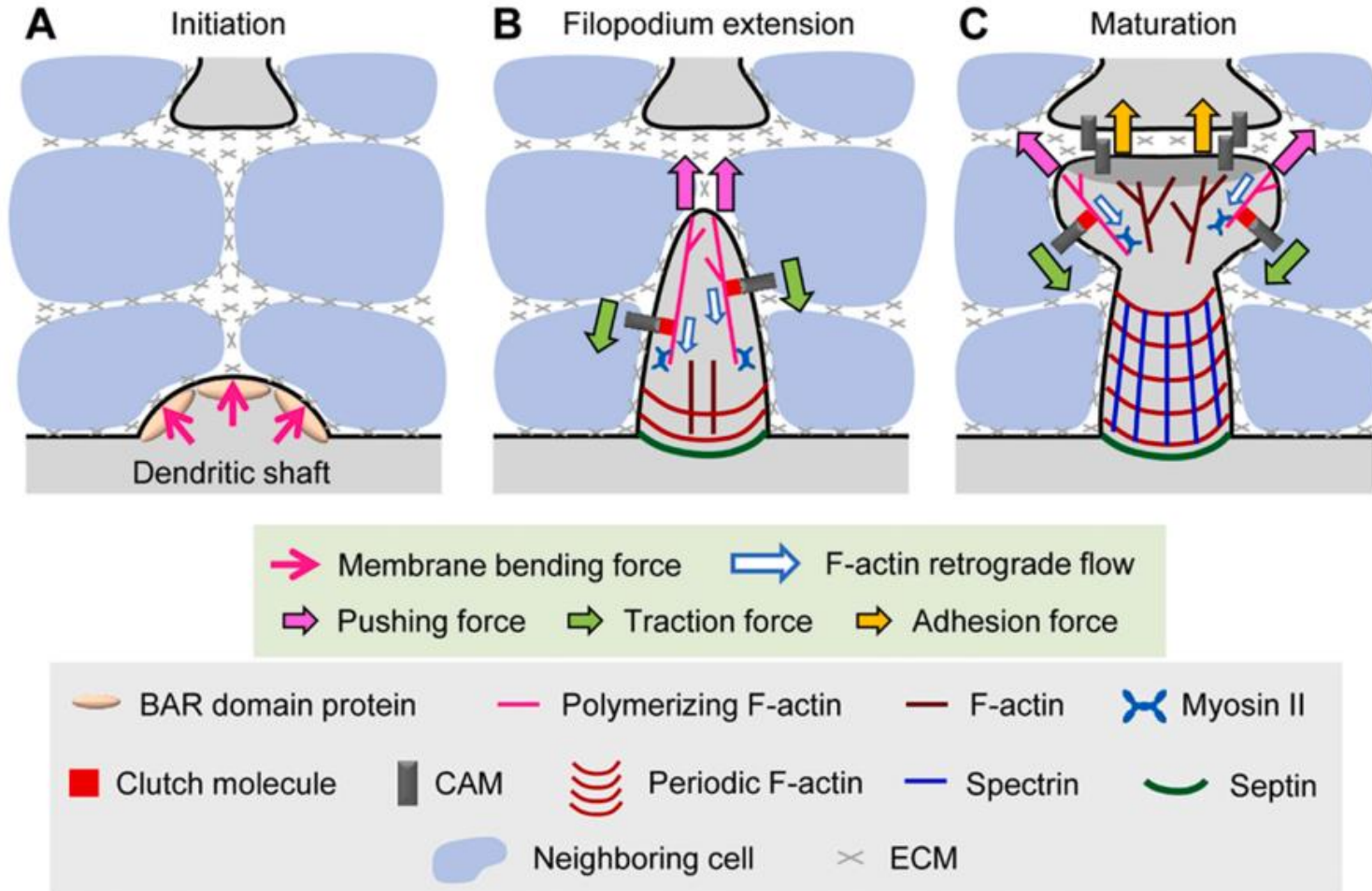
- Dendritic spines are small, actin-rich, bulbous protrusions on neuron dendrites.
- They are the primary sites of excitatory synaptic input in the brain.
- Spines contain receptors and signalling machinery crucial for synaptic transmission.
- They are dynamic structures, changing with learning and neural activity.
- Alterations in spine shape or density are linked to neurological disorders.

Types of dendritic spines



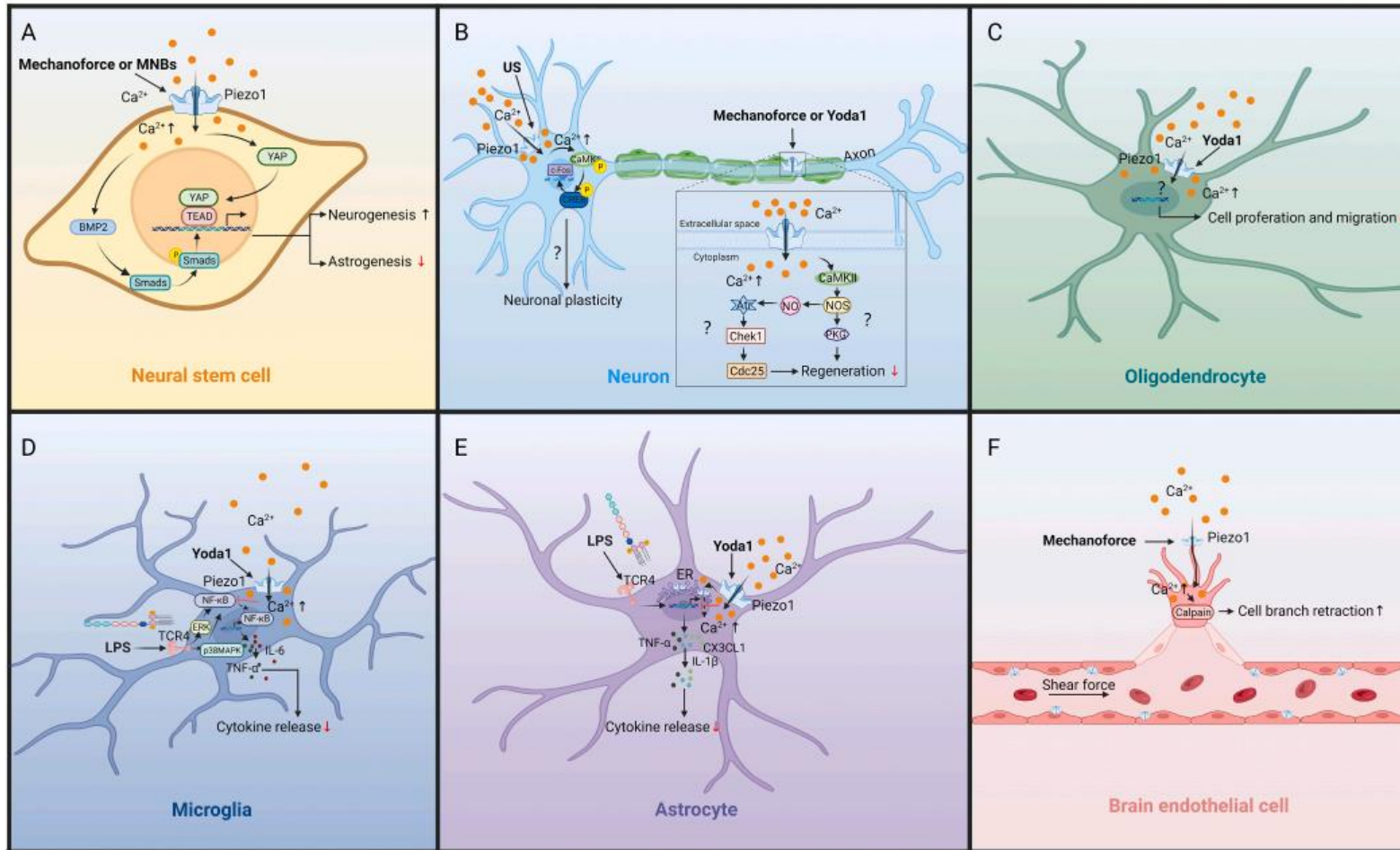
Mechanical forces in dendritic spines

From the initiation to the maturation, the dendritic spines are guided by mechanical forces



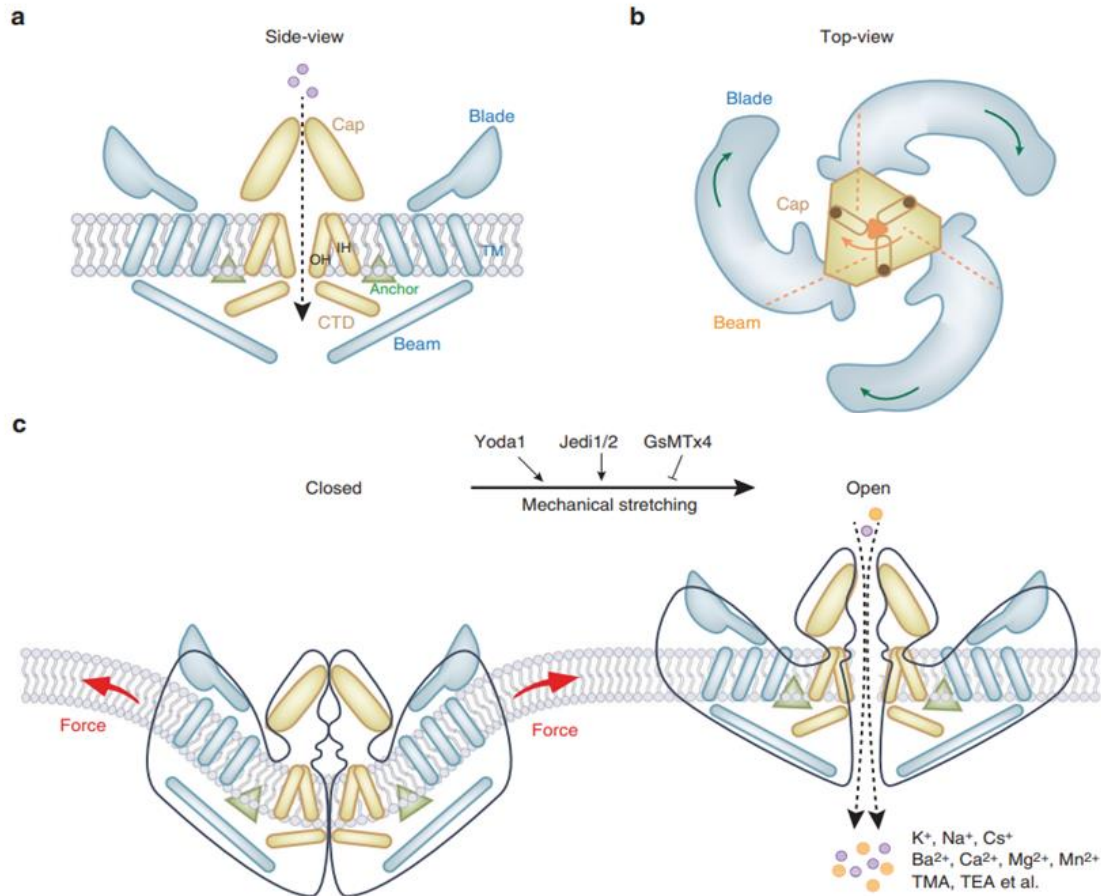
Piezo1 is a mechanical force sensor

Mechanosensitive Piezo1 channel in physiology and pathophysiology of the central nervous system



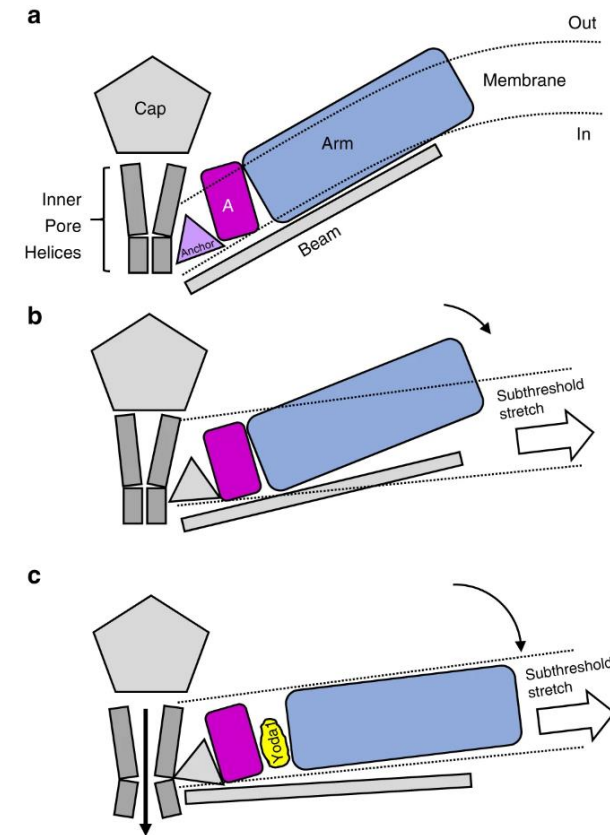
Piezo1 is a mechanical force sensor

Piezo1 is a mechanically activated cation channel



Qin et al., 2021

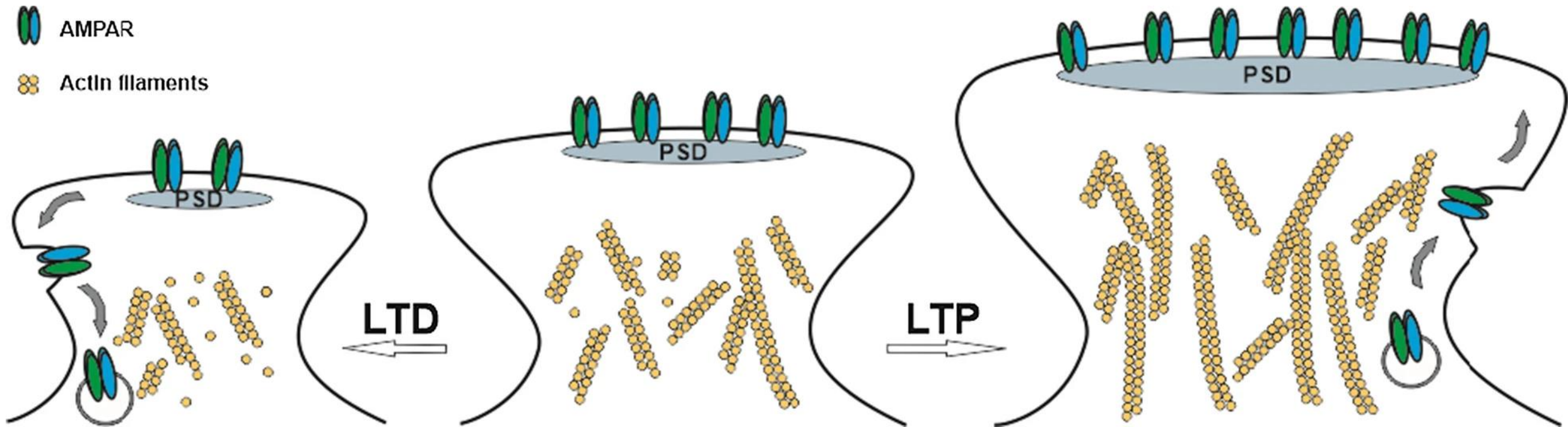
Piezo1 activity can be manipulated pharmacologically



Botello-Smith et al., 2019

Dendritic spines and long-term potentiation

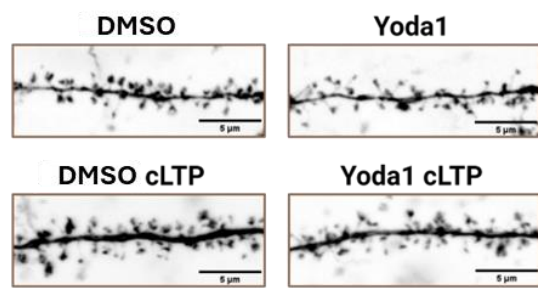
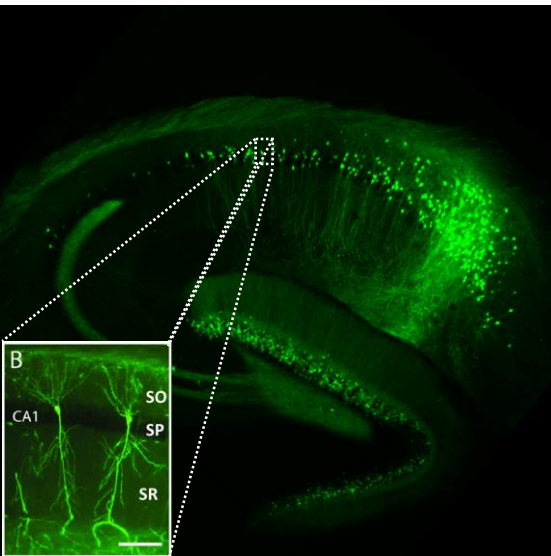
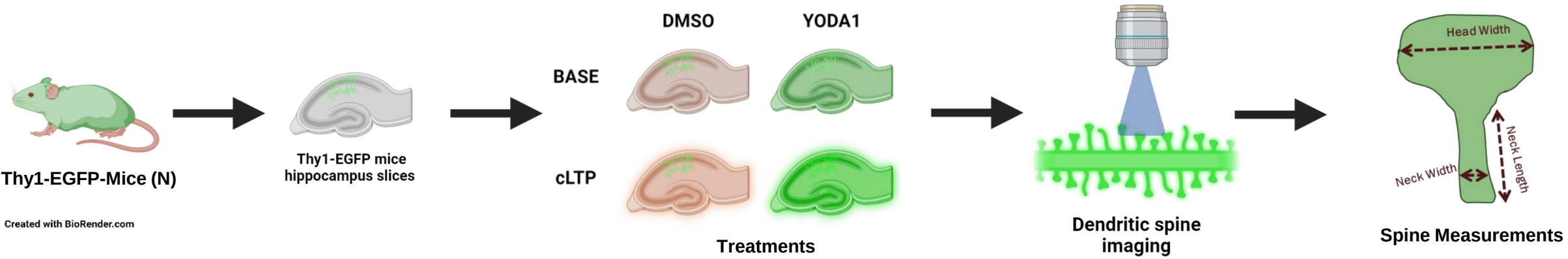
Long-term potentiation (LTP) is a key mechanism of synaptic plasticity that strengthens synaptic connections after repeated stimulation



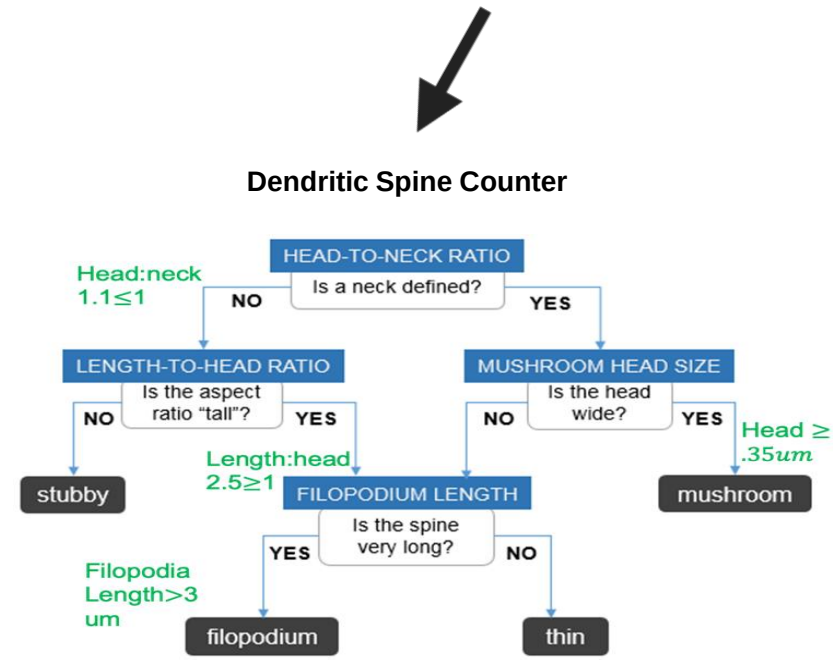
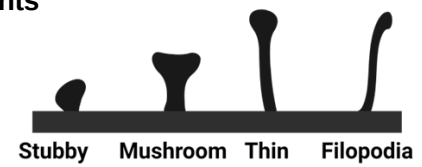
Scientific question

How does Piezo1 activation modulate dendritic spine structural plasticity during chemically induced long-term potentiation in hippocampal neurons?

Workflow – Sample processing



Dendritic Spine Segments

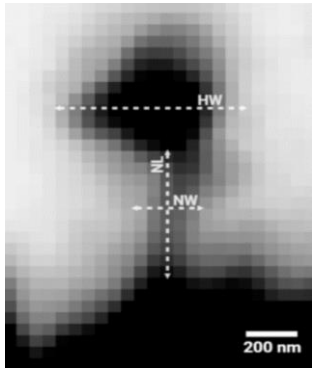
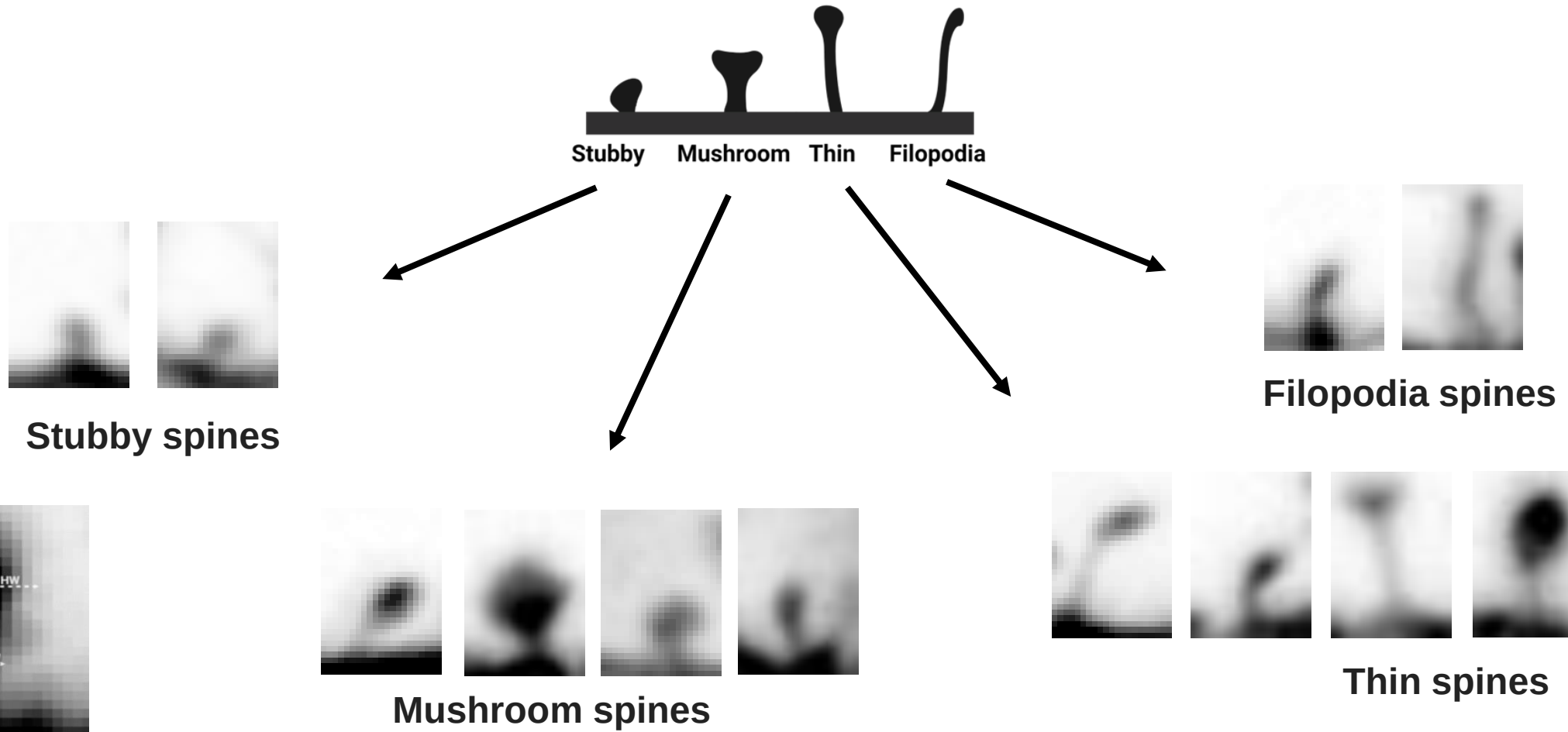


Blinding

- Every statistical test assumes random sampling.
- Blinding can be performed before image analysis or even better, before imaging.

Original Name (.tif)	New Name (.tif)
1	zq7lKgr6
2	hioYE39s
3	qvgZk4F8
4	wS7wE9jz
5	9mGnUeyV

Examples



Workflow – Data arrangement

Unique, used for identification

Dendrite Segment ID	Mouse	Slice	DMSO	Yoda1	cLTP	Spine ID	Classification	Neck Length (μm)	Neck Width (μm)	Head Width (μm)
26	1	3	1	0	0	1393	filopodia	1.402	0.157	0.087
111	2	18	1	1	0	3503	filopodia	0.632	0.089	0.052
195	3	36	1	0	1	1019	filopodia	0.698	0.139	0.133
200	1	38	1	1	1	3067	filopodia	1.074	0.102	0.096
1	1	1	1	0	0	1985	mushroom	0.401	0.107	0.402
80	3	13	1	1	0	955	mushroom	0.247	0.058	0.324
147	1	26	1	0	1	218	mushroom	0.117	0.111	0.408
230	1	43	1	1	1	1542	mushroom	0.215	0.192	0.326
6	2	1	1	0	0	2994	stubby	0.093	0.128	0.289
80	1	13	1	1	0	969	stubby	0.023	0.223	0.213
144	1	26	1	0	1	3323	stubby	0.037	0.196	0.228
198	1	38	1	1	1	3024	stubby	0.088	0.231	0.299
6	2	1	1	0	0	2989	thin	0.793	0.082	0.179
80	1	13	1	1	0	957	thin	0.404	0.057	0.365
144	1	26	1	0	1	3308	thin	0.847	0.109	0.215
279	3	52	1	1	1	3306	thin	0.456	0.098	0.164

Workflow – Data arrangement

Unique, used for structure level
identification and random effect

Dendrite Segment ID	Mouse	Slice	DMSO	Yoda1	cLTP	Spine ID	Classification	Neck Length (μm)	Neck Width (μm)	Head Width (μm)
26	1	3	1	0	0	1393	filopodia	1.402	0.157	0.087
111	2	18	1	1	0	3503	filopodia	0.632	0.089	0.052
195	3	36	1	0	1	1019	filopodia	0.698	0.139	0.133
200	1	38	1	1	1	3067	filopodia	1.074	0.102	0.096
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144	1	26	1	0	1	3308	thin	0.847	0.109	0.215
279	3	52	1	1	1	3306	thin	0.456	0.098	0.164

Workflow – Data arrangement

Works as yes/no for drug treatments
Used in the model as independent variable

Dendrite Segment ID	Mouse	Slice	DMSO	Yoda1	cLTP	Spine ID	Classification	Neck Length (μm)	Neck Width (μm)	Head Width (μm)
26	1	3	1	0	0	1393	filopodia	1.402	0.157	0.087
111	2	18	1	1	0	3503	filopodia	0.632	0.089	0.052
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144	1	26	1	0	1	3308	thin	0.847	0.109	0.215
279	3	52	1	1	1	3306	thin	0.456	0.098	0.164

Workflow – Data arrangement

Unique, used for spine identification

Dendrite Segment ID	Mouse	Slice	DMSO	Yoda1	cLTP	Spine ID	Classification	Neck Length (μm)	Neck Width (μm)	Head Width (μm)
26	1	3	1	0	0	1393	filopodia	1.402	0.157	0.087
111	2	18	1	1	0	3503	filopodia	0.632	0.089	0.052
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279	3	52	1	1	1	3306	thin	0.456	0.098	0.164

Workflow – Data arrangement

Classification from analysis software
Used to filter the data

Dendrite Segment ID	Mouse	Slice	DMSO	Yoda1	cLTP	Spine ID	Classification	Neck Length (μm)	Neck Width (μm)	Head Width (μm)
26	1	3	1	0	0	1393	filopodia	1.402	0.157	0.087
111	2	18	1	1	0	3503	filopodia	0.632	0.089	0.052
195	3	36	1	0	1	1019	filopodia	0.698	0.139	0.133
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279	3	52	1	1	1	3306	thin	0.456	0.098	0.164

Workflow – Data arrangement

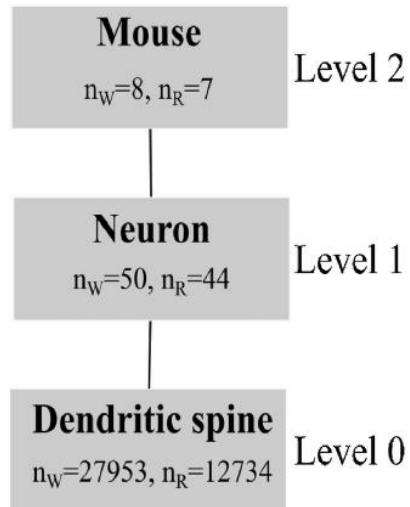
Parameters measured from analysis software
Used in the model as dependent variable

Dendrite Segment ID	Mouse	Slice	DMSO	Yoda1	cLTP	Spine ID	Classification	Neck Length (μm)	Neck Width (μm)	Head Width (μm)
26	1	3	1	0	0	1393	filopodia	1.402	0.157	0.087
111	2	18	1	1	0	3503	filopodia	0.632	0.089	0.052
195	3	36	1	0	1	1019	filopodia	0.698	0.139	0.133
200	1	38	1	1	1	3067	filopodia	1.074	0.102	0.096
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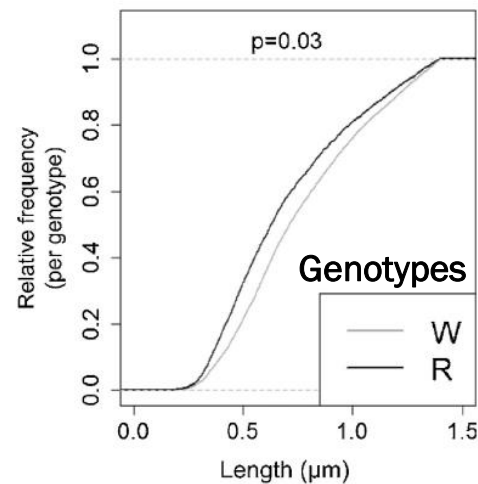
Spine analysis

Data analysis – Using mixed-effects models to deal with multilevel-structured data

A



B



Dendritic Spine Parameters = dependent variable
the continuous measurement to be modelled

Genotype = independent variable
main/fixed effects tested for their influence on the dependent variable

(1|Neuron), (1|Mouse) = random effect
*allowing for individual variability in the baseline level of the dendritic spine parameter among different structure levels

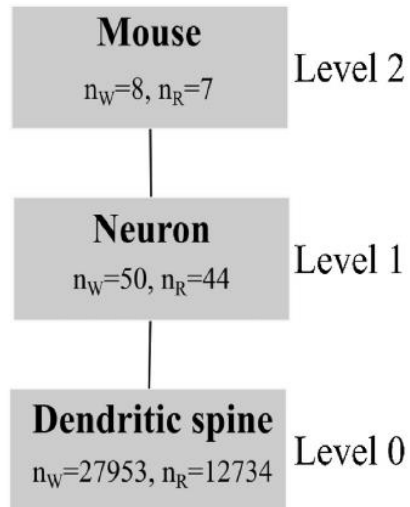
Dendritic Spine Parameter ~ Genotype + (1|Neuron) + (1|Mouse)

Paternoster et al., 2018

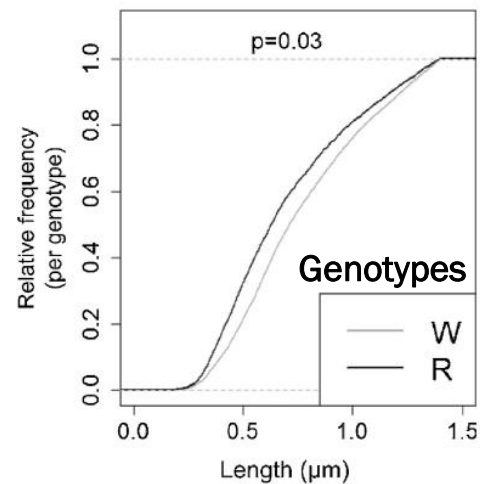
Spine analysis

Data analysis – Using mixed-effects models to deal with multilevel-structured data

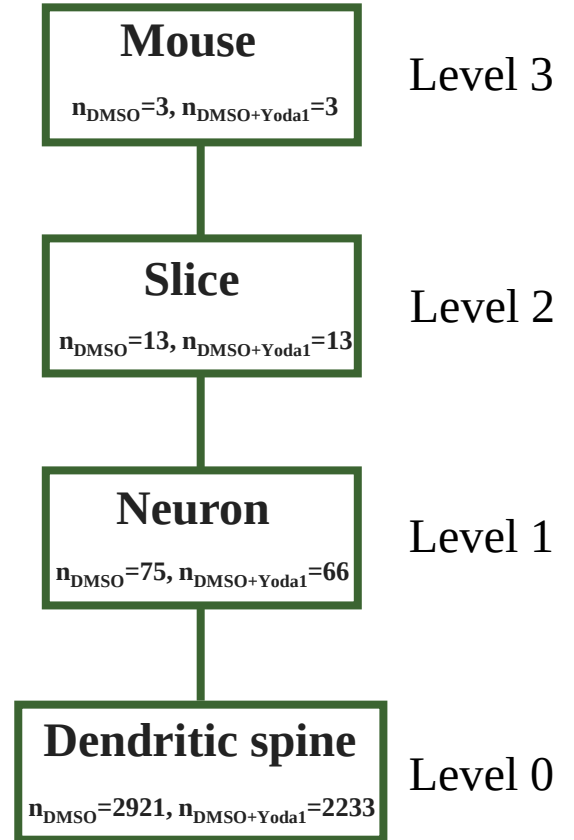
A



B



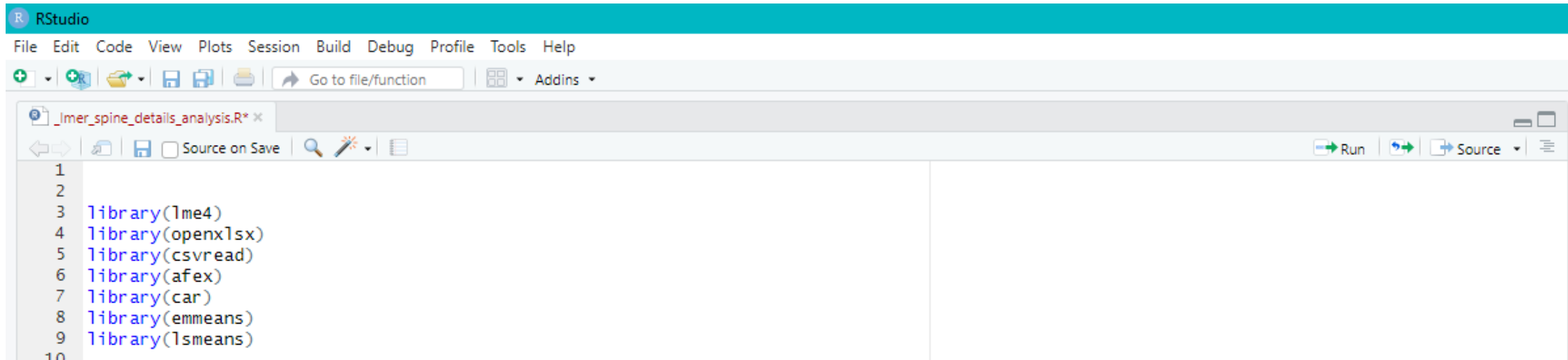
Dendritic Spine Parameter $\sim$ Genotype + (1 | Neuron) + (1 | Mouse)



Paternoster et al., 2018

Workflow – Analysis using lme4 (in R)

Loading the libraries



The screenshot shows the RStudio environment. The top menu bar includes File, Edit, Code, View, Plots, Session, Build, Debug, Profile, Tools, and Help. Below the menu is a toolbar with icons for file operations and a search bar labeled 'Go to file/function'. The main editor window displays a script file named '_lmer_spine_details_analysis.R*'. The script contains the following code:

```
1  
2  
3 library(lme4)  
4 library(openxlsx)  
5 library(csvread)  
6 library(afex)  
7 library(car)  
8 library(emmeans)  
9 library(lsmmeans)  
10
```

The right side of the RStudio interface shows the 'Run' and 'Source' buttons, along with a 'Source on Save' checkbox.

Workflow – Analysis using lme4 (in R)

Reading and storing the data from table

```
11 raw_data <- read.csv("merged_file_python_average.csv")
12 mushroom_data <- subset(raw_data, Classification == "mushroom")
13 thin_data <- subset(raw_data, Classification == "thin")
14 stubby_data <- subset(raw_data, Classification == "stubby")
15 filopodia_data <- subset(raw_data, Classification == "filopodia")
```

Defining the dependent variable

```
17 dependent_variable <- filopodia_data$Neck_Length
```

Define the model/models

```
18 model1 <- dependent_variable ~ 1 + Yoda1*cLTP + (1|Mouse) + (1|slice)
19 model2 <- dependent_variable ~ 1 + Yoda1 + cLTP + (1|Mouse) + (1|slice)
20
21 m1 <- lmer(model1, data = filopodia_data, REML = TRUE)
22 m2 <- lmer(model2, data = filopodia_data, REML = TRUE)
```

Question Model 1: Does the effect of Yoda1 treatment on the dependent variable depend on the presence of chemical LTP?
Does combining treatments produce a different effect than either alone?

Question Model 2: Do Yoda1 treatment and chemical LTP independently affect the dependent variable?

Workflow – Analysis using lme4 (in R)

Fitting, printing and results

```
24 r_squared_m1 <- cor(dependent_variable, predict(m1))^2
25 r_squared_m2 <- cor(dependent_variable, predict(m2))^2
26
27 anova_m1 <- anova(m1)
28 anova_m2 <- anova(m2)
29
30 print(summary(m1))
31 print(summary(m2))
32
33 print(anova_m1)
34 print(anova_m2)
35
36 print(r_squared_m1)
37 print(r_squared_m2)|
```

Workflow – Analysis using lme4 (in R)

summary(m1)

```
> print(summary(m1))
Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
Formula: model1
Data: filopodia_data

REML criterion at convergence: -106.3

Scaled residuals:
    Min       1Q   Median       3Q      Max
-1.6345 -0.5169 -0.1499  0.4056  2.5644

Random effects:
 Groups   Name      Variance Std.Dev.
 Slice    (Intercept) 0.000000 0.00000
 Mouse    (Intercept) 0.000000 0.00000
 Residual                    0.001278 0.03575
Number of obs: 34, groups:  slice, 22; Mouse, 3

Fixed effects:
              Estimate Std. Error      df t value Pr(>|t|)
(Intercept)  0.092286   0.013511 30.000000   6.831 1.41e-07 ***
Yoda1        0.001048   0.018014 30.000000   0.058  0.95401
cLTP         0.064143   0.019107 30.000000   3.357  0.00215 **
Yoda1:cLTP   -0.084749   0.024964 30.000000  -3.395  0.00195 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
              (Intr) Yoda1  cLTP
Yoda1        -0.750
cLTP         -0.707  0.530
Yoda1:cLTP    0.541 -0.722 -0.765
```

Workflow – Analysis using lme4 (in R)

Comparing the models

```
39 print(anova(m1, m2))
```

	npar	AIC	BIC	logLik	deviance	Chisq	Df	Pr(>Chisq)
m2	6	48.784	57.943	-18.392	36.784			
m1	7	49.900	60.585	-17.950	35.900	0.8842	1	0.3471

npar – Number of parameters in the model (fixed + random + residual variance). m2: 6, m1: 7 (m1 adds one parameter — the interaction)

AIC – Akaike Information Criterion: a measure of model fit penalized by complexity. The lower the AIC, the better trade-off between fit and complexity. $\Delta AIC < 2$ = minimal differences.

BIC – Bayesian Information Criterion: like AIC, but penalizes complexity more heavily. Again, lower is better. $\Delta BIC < 2$ = minimal differences.

logLik – Log-likelihood of the model (higher is better).

deviance – $-2 \times \text{log-likelihood}$. Lower deviance means a better fit.

Chisq – Chi-square statistic from the likelihood ratio test (LRT). This tests whether the more complex model fits significantly better than the other.

Df – Degrees of freedom: difference in number of parameters ($7 - 6 = 1$).

Pr(>Chisq) – p-value of the test. The value 0.3471 means that the difference is not statistically significant.

You need to find the best model while keeping in mind your scientific question.

Workflow – Analysis using lme4 (in R)

That's how easy it is!

The screenshot displays the RStudio interface with an R script and its execution output.

R Script (lmer_spine_details_analysis.R):

```
1 library(lme4)
2 library(openxlsx)
3 library(csvread)
4 library(afex)
5 library(car)
6 library(emmeans)
7 library(lsmmeans)
8
9 raw_data <- read.csv("merged_file_python_average.csv")
10 mushroom_data <- subset(raw_data, classification == "mushroom")
11 thin_data <- subset(raw_data, classification == "thin")
12 stubby_data <- subset(raw_data, classification == "stubby")
13 filopodia_data <- subset(raw_data, classification == "filopodia")
14
15 dependent_variable <- filopodia_data$Neck_Length
16 model11 <- dependent_variable ~ 1 + Yoda1*CLTP + (1|Mouse) + (1|Slice)
17 model12 <- dependent_variable ~ 1 + Yoda1 + CLTP + (1|Mouse) + (1|Slice)
18
19 m1 <- lmer(model11, data = filopodia_data, REML = TRUE)
20 m2 <- lmer(model12, data = filopodia_data, REML = TRUE)
21
22 r_squared_m1 <- cor(dependent_variable, predict(m1))^2
23 r_squared_m2 <- cor(dependent_variable, predict(m2))^2
24
25 anova_m1 <- anova(m1)
26 anova_m2 <- anova(m2)
27
28 print(summary(m1))
29 print(summary(m2))
30
31 print(anova_m1)
32 print(anova_m2)
33
34 print(r_squared_m1)
35 print(r_squared_m2)
36
37 print(anova(m1, m2))
38
39
40
```

Environment Panel:

Object	Class	Size
Global Environment		
raw_data	data.frame	34 obs. of 11 variables
mushroom_data	data.frame	5154 obs. of 24 variables
thin_data	data.frame	Formal class 'lmerModLmerTest'
stubby_data	data.frame	Formal class 'lmerModLmerTest'
thin_data	data.frame	Formal class 'lmerMod'
mushroom_data	data.frame	658 obs. of 11 variables
raw_data	data.frame	3466 obs. of 11 variables
stubby_data	data.frame	81 obs. of 11 variables
thin_data	data.frame	2693 obs. of 11 variables

Files Panel:

Name	Size	Modified
..		
lmer_spine_details_analysis.R	1.1 KB	Jun 17, 2025, 9:59 AM
.RData	535.2 KB	May 23, 2024, 6:46 PM
.Rhistory	14.6 KB	May 23, 2024, 6:46 PM
All_Figures.pdf	2.5 MB	May 23, 2024, 6:47 PM
copy_csv.py	2 KB	May 6, 2024, 1:20 AM
Database_Spine_Details_R.csv	133.3 KB	May 6, 2024, 6:46 PM
Dendrite_Spine_Details_Blinded_Tables		
Dendrite_Spine_Summary_Table_Blinded.csv	8.3 KB	May 6, 2024, 12:20 AM
Dendrite_Spine_Summary_Table_De-Blinded_R_AVG.xlsx	28.6 KB	May 7, 2024, 8:50 AM
Dendrite_Spine_Summary_Table_De-Blinded_R.csv	6.8 KB	May 6, 2024, 12:44 AM
Dendrite_Spine_Summary_Table_De-Blinded.csv	7.7 KB	May 6, 2024, 6:07 PM
extract_average_values.py	1.6 KB	May 23, 2024, 4:56 PM
grouped_output_with_averages_filopodia_2.csv	653 B	May 23, 2024, 4:56 PM
grouped_output_with_averages_filopodia.csv	676 B	May 23, 2024, 4:07 PM
grouped_output_with_averages_mushroom.csv	895 B	May 23, 2024, 3:52 PM
grouped_output_with_averages_stubby.csv	748 B	May 23, 2024, 4:07 PM
grouped_output_with_averages_thin.csv	895 B	May 23, 2024, 4:06 PM
lmer_spine_density_analysis.R	639 B	May 7, 2024, 9:15 AM
lmer_spine_density_analysis.R	2.6 KB	May 6, 2024, 6:41 PM
Merge_Spine_Database.py	506 B	May 6, 2024, 6:37 PM
merged_file_mushroom.csv	5.1 KB	May 16, 2024, 8:16 PM
merged_file_python_average.csv	148.3 KB	May 23, 2024, 4:55 PM
merged_file_thin.csv	21.7 KB	May 16, 2024, 8:31 PM
merged_file.csv	328.8 KB	May 6, 2024, 7:05 PM
Plugin_Dendrite_Segments_Blinded		
Spine_Analysis_April_2024-KEY.xlsx	10.5 KB	May 6, 2024, 1:08 AM

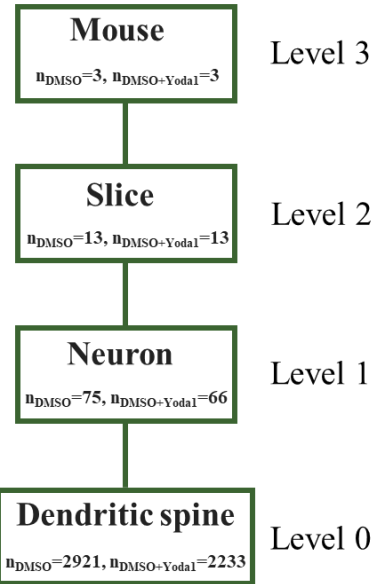
Console Output:

```
R 4.3.1 - C:/Users/user/OneDrive/Desktop/Spine Analysis/Spines_April_2024/De-blinded_Analysis/
> plot(predict(m1), dependent_variable, main = "total spine density", xlab = "fitted values", ylab = "residuals")
>
> print(anova(m1, m2))
refitting model(s) with ML (instead of REML)
data: filopodia_data
Models:
m2: model12
m1: model11
      npar  AIC      BIC logLik deviance Chisq Df Pr(>Chisq)
m2      6 48.784 57.943 -18.392   36.784      NA  NA
m1      7 49.900 60.585 -17.950   35.900 0.8842  1    0.3471
>
```

Why LME?

To account for hierarchical or nested data in your experiment.

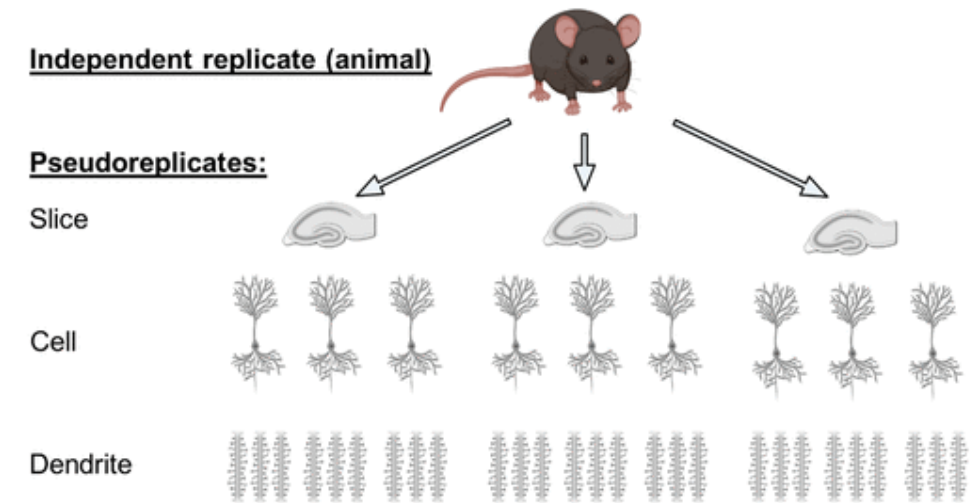
	Treatment				N Mice	N Slice (350 µm)	N Slice (100 µm)	N Neurons	N Dendritic Segment	Total N Dendritic Segments
Group	DMSO	Yoda1	E64	cLTP						
DMSO Base	+	-	-	-	6	2	3	3	1	108
DMSO cLTP	+	-	-	+	6	2	3	3	1	108
Yoda1 Base	+	+	-	-	6	2	3	3	1	108
Yoda1 cLTP	+	+	-	+	6	2	3	3	1	108
E64 Base	+	+	+	-	6	2	3	3	1	108
E64 cLTP	+	+	+	+	6	2	3	3	1	108
Total										648



Why LME?

To prevent pseudoreplication.

- In simple linear models each spine would be treated as an independent observation, if the data is not averaged per each structure level.
- If the data is averaged, then you will lose a lot of information about spine variability.
- Mixed models avoid this by modeling variability within and between higher-level units (like animals, slices and dendrites).



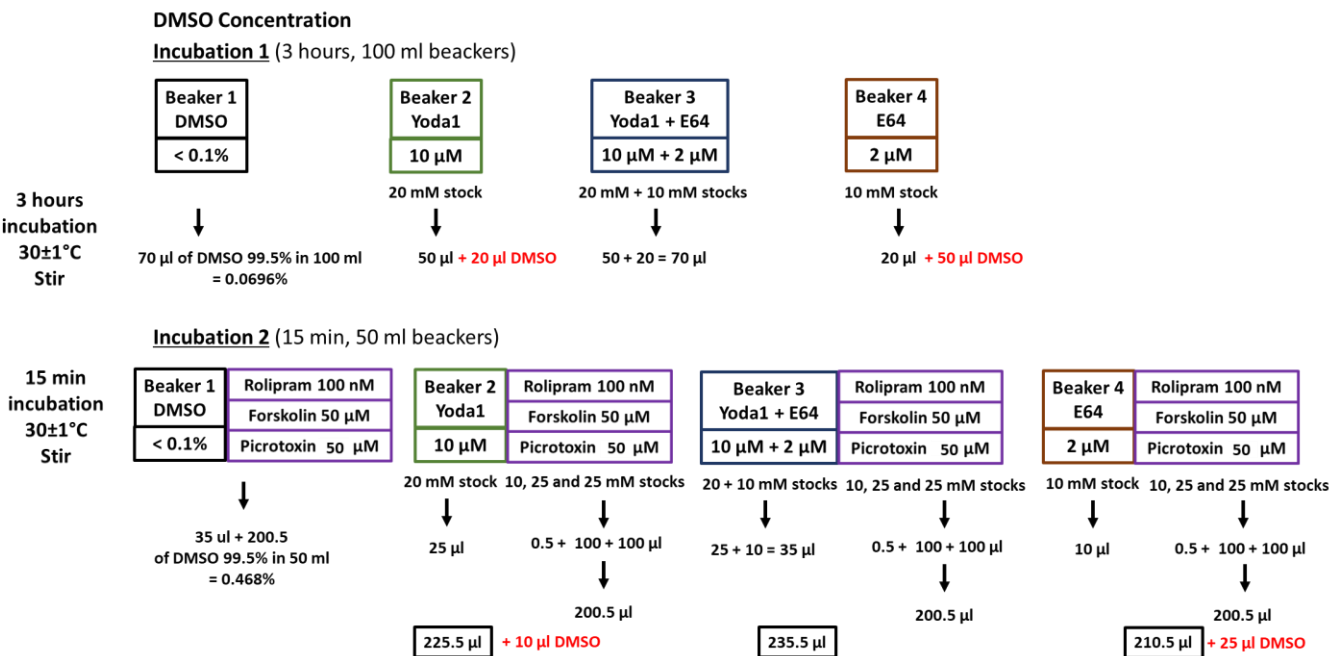
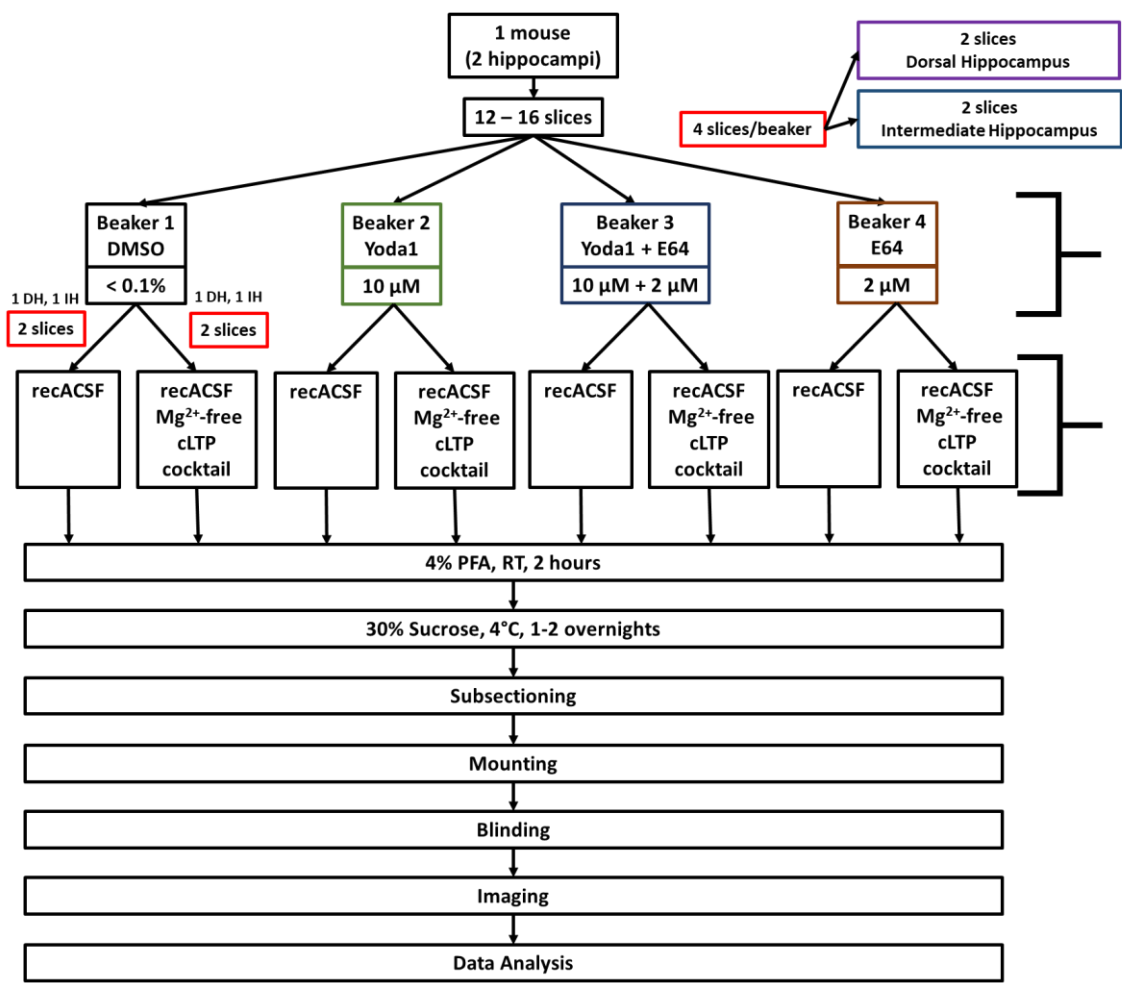
Student's t-test (assuming independent replicates)

$$t = \frac{m - \mu}{s/\sqrt{n}}$$

m = Mean
 μ = theoretical value (control mean)
s = pooled standard deviation
n = degrees of freedom

Why LME?

To get the most information as possible from your hard-earned data



Conclusions



Journal of Neuroscience Methods

Volume 296, 15 February 2018, Pages 93-98



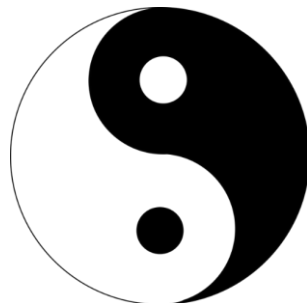
Short communication

The importance of data structure in statistical analysis of dendritic spine morphology

Veerle Paternoster^{a b c d}  , Anto P. Rajkumar^{a b d e f}  , Jens Randel Nyengaard^g  ,
Anders Dupont Borglum^{a b d}  , Jakob Grove^{a b d h i}  , Jane Hvarregaard Christensen^{a b d i}  

Highlights

- Data collection for dendritic spine morphology often results in multilevel data structure.
- The need for implementation of data structure in statistical models is highlighted.
- The use of mixed-effects models to study dendritic spine morphology is proposed.
- Mixed-effects models are superior to commonly used statistical methods for studying dendritic spine morphology.



Current Research in Neurobiology

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Unnecessary reliance on multilevel modelling to analyse nested data in neuroscience: When a traditional summary-statistics approach suffices

Carolyn Beth McNabb^a  , Kou Murayama^{a b c}  

Highlights

- Multilevel modelling is not superior to summary statistics when clusters are balanced.
- Summary statistics provides benefits when singular fit errors occur.
- Previous reports of superior performance with multilevel modelling are not supported.

+ Very conservative, average